



OPEN Association between visceral adipose tissue measured by deep neural network architecture and chronic kidney disease

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The relationship between abdominal body composition and chronic kidney disease (CKD) is well-documented. In this study, we aimed to investigate the association between CKD and abdominal fat volume assessed using deep neural network architecture. This study used the health check-up data of 14,105 patients with available computed tomography (CT) images of the abdomen in a Korean population. The volumes of body segments, including visceral adipose tissue (VAT) and subcutaneous adipose tissue (SAT), were measured using an artificial intelligence (AI)-based image analysis software. Of the 14,105 participants, the prevalence of CKD was 2.3% in males and 1.3% in females. In the multivariable analysis, the volumes of VAT were significantly associated with an increased risk of CKD in both males (odds ratio [OR], 1.25, 95% confidence interval [CI], 1.18–1.33, $P < 0.001$) and females (OR, 1.18, 95% CI, 1.10–1.27, $P < 0.001$). The volumes of SAT were significantly associated with a decreased risk of CKD in females (OR, 0.77, 95% CI, 0.71–0.83, $P < 0.001$) and an increased risk of CKD in males (OR, 1.09, 95% CI, 1.03–1.16, $P < 0.001$). The VAT/SAT volume ratio was significantly associated with an increased risk of CKD in males (OR, 1.09; 95% CI, 1.04–1.14, $P < 0.001$). Abdominal fat volumes were independently associated with CKD risk, showing positive associations of VAT in both sexes and inverse associations of SAT in women. These findings highlight the importance of considering sex-specific abdominal fat distribution in CKD risk evaluation. The use of AI-based analysis of abdominal CT images may help improve early detection and risk stratification of CKD.

Keywords Artificial intelligence, Body composition, Kidney disease, Adipose

Chronic kidney disease (CKD) is a major global health concern linked to higher mortality rates, rising socioeconomic costs, and a burden on health-related quality of life^{1,2}. The global prevalence of CKD is estimated to be around 9%, affecting nearly 700 million people worldwide, and continues to increase due to population aging and the growing incidence of diabetes, hypertension, and obesity^{3–5}. Among these factors, obesity is a key determinant of CKD onset, kidney function decline, and mortality risk^{6,7}.

Although body mass index (BMI) remains the most widely used tool for assessing obesity in clinical practice, and epidemiologic studies have demonstrated a link between high BMI and CKD incidence⁸, BMI does not always accurately reflect excess body fat and may be influenced by CKD-related change⁹. Computed tomography (CT) provides high tissue resolution, allowing clear differentiation between visceral and subcutaneous fat. Cross-sectional CT scans have been utilized for precise measurement of body fat composition¹⁰. Recently, deep-learning algorithms have been applied to medical imaging, enabling automatic volumetric segmentation of body components in CT images^{11,12}. This advancement allows for the extraction of three-dimensional (3D) volumetric data of body composition from clinical CT scans, potentially offering advantages over traditional two-dimensional manual segmentation. Unlike single-slice planimetric measurements, a deep learning-based approach to measure the entire volume of abdominal adipose tissue, can provide a more comprehensive and accurate representation of an individual's visceral adiposity. As demonstrated in recent literature, volumetric

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analysis of abdominal fat using CT is a powerful predictor for the prevalence and incidence of metabolic diseases¹³.

Sex-related differences in adipose tissue distribution and metabolic regulation may influence the relationship between abdominal fat and kidney function. Men generally accumulate more visceral adipose tissue (VAT), whereas women tend to have greater subcutaneous adipose tissue (SAT), which may exert protective metabolic effects^{14,15}. Given these differences, the present study aimed to examine the association between volumetrically measured abdominal fat and CKD risk in a Korean population, with particular attention to potential sex-specific patterns.

Methods

Study participants and study design

This retrospective cross-sectional study consisted of adults who voluntarily obtained abdominal CT scans for health check-ups at Seoul National University Hospital Healthcare System Gangnam Center from January 1, 2011, to September 30, 2012. Among all individuals, we excluded participants with missing estimated glomerular filtration rate (eGFR) data ($n = 11$), those younger than 40 years ($n = 1174$), and those older than 80 years ($n = 34$). We further excluded individuals with a known history of malignancy at baseline ($n = 6$). After these exclusions, a total of 14,105 participants aged 40–80 years, comprising 8510 males and 5595 females, were included in the final analysis.

This study was approved by the Institutional Review Board of Seoul National University Hospital (No.2102-176-1200). Due to the retrospective nature of the study, Institutional Review Board of Seoul National University Hospital waived the need of obtaining informed consent. This study was conducted according to the Declaration of Helsinki.

Clinical and laboratory evaluation

A questionnaire was used to obtain information regarding the patients' medical history and smoking status. BMI was calculated as weight (kg)/height (m²), and waist circumference (WC) was measured at the midpoint between the lower costal margin and the iliac crest. Blood samples were collected after a 12-h overnight fast. Triglycerides, total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, fasting plasma glucose, glycosylated hemoglobin (HbA1c), creatinine, and uric acid were utilized as biochemical test data. Participants were classified as having diabetes mellitus if their fasting glucose levels were ≥ 126 mg/dL, if their HbA1c levels were $\geq 6.5\%$, or if they had ever been diagnosed with diabetes by a physician or were taking anti-diabetic medications. Hypertension was defined as a systolic blood pressure of ≥ 140 mmHg and a diastolic blood pressure of ≥ 90 mmHg, or treatment with anti-hypertensive agents. Albuminuria was assessed using a spot urine albumin-to-creatinine ratio. CKD was defined as an eGFR < 60 mL/min/1.73 m², which was calculated using the equation below.

$$\text{eGFR} = 175 \times (\text{SCr})^{-1.154} \times (\text{age})^{-0.203} \times 0.742 \text{ [if female]} \times 1.212 \text{ [if Black]}$$

Abbreviations/Units

eGFR (estimated glomerular filtration rate) = mL/min/1.73 m²

SCr (standardized serum creatinine) = mg/dL

age = years

Body composition analysis

Body composition was assessed using an artificial intelligence-based CT analysis software (DeepCatch, version 1.0.0.0; MEDICALIP Co., Ltd., Seoul, Korea). The software uses a fixed U-Net-based deep learning architecture and a predefined segmentation pipeline, as previously validated^{16,17}, where the complete model architecture, feature extraction procedures, and validation performance metrics are detailed. All internal preprocessing steps, segmentation thresholds, and post-processing procedures are embedded within the software and are not user-configurable. Accordingly, all measurements in this study were generated using the default, non-modifiable settings to ensure consistency and reproducibility. This software automatically selects images of the upper abdomen between the lower border of the rib cage and the top of the iliac crest. It differentiates and quantifies the volumes of the body segments (Fig. 1). The results were provided as the volume of each body segment. Among the volumetric measures of body components, the volumes of VAT and SAT were evaluated for association with CKD. The volumetric variables collected in millimeters (mm) had excessively high data units. Thus, they were converted to cubic centimeters (cm³) by dividing them by 1000.

Statistical analysis

Continuous variables were expressed as medians [interquartile ranges] and compared between groups using the Mann–Whitney U test. Categorical variables were presented as numbers (percentages) and compared using the chi-square test. Normality was assessed using the Shapiro–Wilk test.

To account for data imbalance between CKD and non-CKD groups, class weighting was applied during model fitting. Categorical variables were converted into dummy indicator variables, and continuous variables were standardized before analysis. Multivariable logistic regression was used to assess the associations between body composition parameters and CKD risk. Covariates were selected based on clinical relevance and prior evidence linking abdominal adiposity to CKD¹⁸. The key assumptions of logistic regression were evaluated. Linearity of the logit was assessed using the Box–Tidwell test and restricted cubic spline analysis (RCS). Multicollinearity was examined by calculating variance inflation factors, all of which were < 2 . Model fit was assessed using

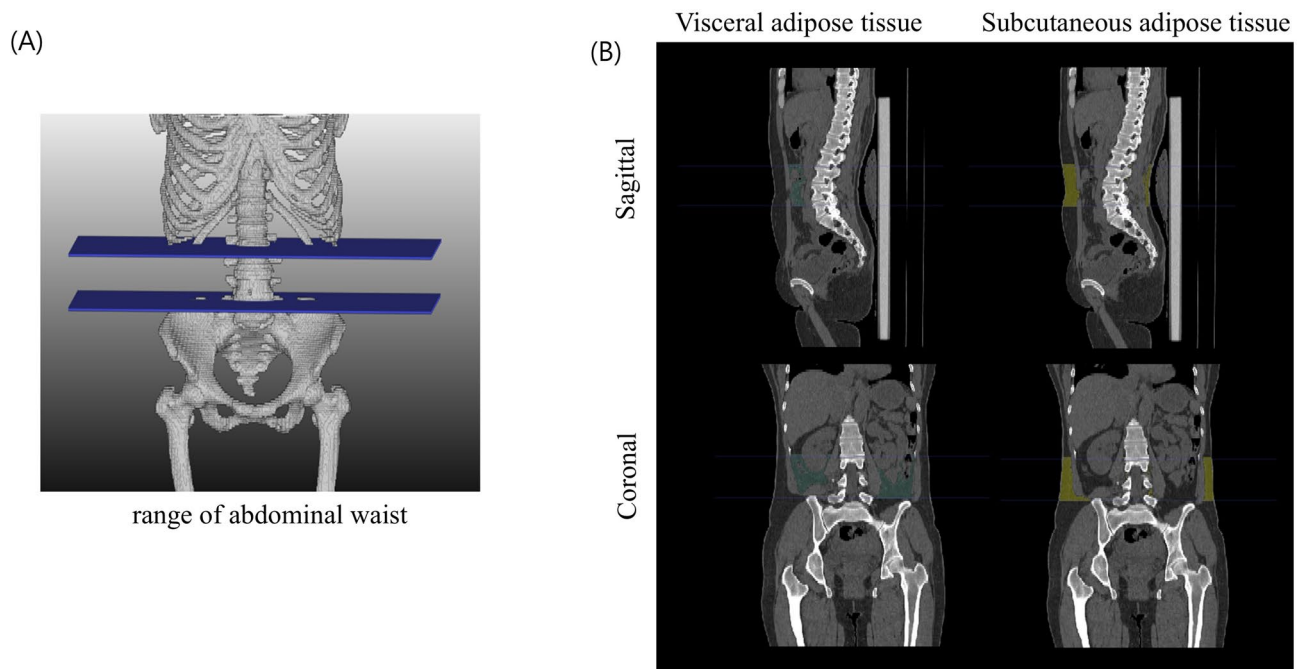


Fig. 1. Automatic labeling of the upper abdomen (A) and body segmentation (B). An artificial intelligence-based computed tomography image analysis software automatically selects images of the upper abdomen between the lower border of the rib cage and the top of the iliac crest. It differentiates and quantifies the volumes of the body segments.

the Hosmer–Lemeshow goodness-of-fit test. All assumptions were satisfied, confirming the adequacy of the model. All statistical analyses were performed using Python version 3.7.6. The result was considered statistically significant if the *p*-value was less than 0.05.

Results

Study population

Of the 14,105 participants included in the study, 8,510 (60.3%) were male. The prevalence of CKD was 2.3% in males and 1.3% in females. Table 1 shows the clinical baseline characteristics of the study population by sex. In both sexes, the proportion of hypertension, diabetes, BMI, WC, and serum levels of HbA1c and uric acid were higher in patients with CKD compared to those without CKD ($P < 0.05$). The volumes of VAT and the VAT/SAT ratios were larger in patients with CKD compared to those without CKD in both males and females.

Association between CKD and body composition parameters

We performed logistic regression analyses to investigate the independent association between body composition parameters and CKD. In the univariate model, the volumes of VAT were significantly associated with an increased risk of CKD in both sexes (males; odds ratio [OR] 1.29, 95% confidence interval [CI], 1.24–1.33, females; OR, 1.52, 95% CI, 1.46–1.58, both $P < 0.001$). The ratio of VAT/SAT was significantly associated with a higher risk of CKD (males; OR, 1.38, 95% CI, 1.33–1.42, females; OR, 1.54, 95% CI, 1.49–1.60, both $P < 0.001$, Table 2).

Table 3 shows the results of the multivariable logistic regression analysis. After adjusting for age, hypertension, and diabetes, the volumes of VAT were significantly associated with an increased risk of CKD in both males (OR, 1.25, 95% CI, 1.18–1.33, $P < 0.001$) and females (OR, 1.18, 95% CI, 1.10–1.27, $P < 0.001$). In the multivariable analysis, the volumes of SAT were significantly associated with a decreased risk of CKD in females (OR, 0.77, 95% CI, 0.71–0.83, $P < 0.001$), while the volumes of SAT were significantly associated with an increased risk of CKD in males (OR, 1.09, 95% CI, 1.03–1.16, $P < 0.001$). The VAT/SAT volume ratio was significantly associated with an increased risk of CKD in males (OR, 1.09; 95% CI, 1.04–1.14, $P < 0.001$).

When the volumes of VAT and SAT were used as categorical variables, the highest quartile of VAT volume was significantly associated with an increased risk of CKD in both sexes (males; OR, 1.77, 95% CI, 1.57–2.00, $P < 0.001$ and females; OR, 1.29, 95% CI, 1.10–1.52, $P < 0.001$). Additionally, the highest quartile of SAT volume was significantly associated with a decreased risk of CKD in females (OR, 0.39, 95% CI, 0.33–0.45, $P < 0.001$), while the highest quartile of SAT volume was significantly associated with an increased risk of CKD in males (OR, 1.29, 95% CI, 1.15–1.45, $P < 0.001$, Table 3). The VAT/SAT volume ratio showed a graded positive association with CKD risk in males and an inverse association in females. These categorical analyses were consistent with the main results using continuous variables, supporting the robustness of the findings.

To assess potential non-linear associations between abdominal fat parameters and CKD risk, we performed RCS analyses. The RCS curves indicated approximately linear relationships between VAT, SAT, and VAT/SAT

Characteristics	Male			Female		
	CKD—No (n = 8,316)	CKD—Yes (n = 194)	P value	CKD—No (n = 5,520)	CKD—Yes (n = 75)	P value
Age (years)	53 [48–59]	61 [55–68]	<0.001	53 [48–59]	62 [57–67]	<0.001
Hypertension (%)	3619 (43.5%)	124 (63.9%)	<0.001	1454 (26.3%)	42 (56.0%)	<0.001
Diabetes (%)	1073 (12.9%)	50 (25.8%)	<0.001	337 (6.1%)	9 (12.0%)	0.062
Antihypertensive medication (%)	1520 (18.3%)	80 (41.2%)	<0.001	589 (10.7%)	26 (34.7%)	<0.001
Antidiabetic medication (%)	490 (5.9%)	30 (15.5%)	<0.001	131 (2.4%)	4 (5.3%)	0.200
Lipid-lowering medication (%)	321 (3.9%)	27 (13.9%)	<0.001	212 (3.8%)	9 (12.0%)	<0.001
Current smoker (%)	1961 (26.3%)	27 (16.2%)	0.004	154 (3.6%)	5 (8.1%)	0.123
Body mass index (kg/m ²)	24 [23–26]	25 [24–26]	0.001	22 [20–24]	23 [21–25]	0.003
Waist circumference (cm)	88 [84–93]	90 [85–95]	<0.001	80 [75–85]	83 [79–88]	<0.001
Fasting glucose (mg/dl)	97 [91–106]	100 [90–114]	0.062	91 [86–98]	92 [86–100]	0.371
Total cholesterol (mg/dl)	192 [169–215]	185 [157–207]	0.003	198 [175–221]	203 [181–232]	0.161
Triglyceride (mg/dl)	107 [75–155]	112 [78–149]	0.561	77 [54–109]	85 [62–124]	0.144
HDL-cholesterol (mg/dl)	47 [42–54]	45 [39–51]	<0.001	55 [48–63]	55 [46–68]	0.820
LDL-cholesterol (mg/dl)	122 [102–143]	116 [94–137]	0.014	120 [101–142]	131 [106–143]	0.124
HbA1c (%)	5.7 [5.5–5.9]	5.8 [5.5–6.2]	<0.001	5.6 [5.4–5.9]	5.8 [5.6–6.0]	<0.001
Uric acid (mg/dl)	6.1 [5.3–6.9]	7.0 [6.1–7.9]	<0.001	4.4 [3.9–5.1]	5.3 [4.8–6.5]	<0.001
eGFR (mL/min/1.73 m ²)	85 [76–94]	56 [53–58]	<0.001	90 [80–101]	58 [52–59]	<0.001
UACR (mg/g)	5.0 [4.0–8.0]	8.5 [4.8–16.8]	<0.001	7.0 [5.0–11.0]	16.0 [8.0–133.0]	0.008
Creatinine (mg/dl)	0.93 [0.85–1.02]	1.31 [1.26–1.37]	<0.001	0.68 [0.62–0.76]	1.00 [0.96–1.07]	<0.001
Body composition						
VAT volume (cm ³)	1,468 [1,049–1,878]	1,652 [1,210–2,061]	<0.001	645 [392–968]	878 [591–1,165]	<0.001
SAT volume (cm ³)	1,267 [1,019–1,557]	1,242 [1,011–1,609]	0.780	1,543 [1,248–1,864]	1,567 [1,283–1,833]	0.753
VAT/SAT volume ratio	1.12 [0.87–1.41]	1.24 [1.00–1.56]	<0.001	0.39 [0.27–0.54]	0.54 [0.37–0.72]	<0.001

Table 1. Baseline characteristics by sex groups. Continuous variables are expressed as medians [interquartile ranges], and categorical variables are presented as numbers (percentages). Values were calculated using available data. *CKD* chronic kidney disease, *HDL* high-density lipoprotein, *LDL* low-density lipoprotein, *HbA1c* glycosylated hemoglobin, *eGFR* estimated glomerular filtration rate, *UACR* urine albumin to creatinine ratio, *VAT* visceral adipose tissue, *SAT* subcutaneous adipose tissue.

Variable	Male		Female	
	OR (95% CI)	P value	OR (95% CI)	P value
Univariable				
VAT volume (per SD)	1.29 (1.24–1.33)	<0.001	1.52 (1.46–1.58)	<0.001
SAT volume (per SD)	0.98 (0.95–1.01)	0.173	0.96 (0.92–1.00)	0.060
VAT/SAT volume	1.38 (1.33–1.42)	<0.001	1.54 (1.49–1.60)	<0.001
Multivariable*				
VAT volume (per SD)	1.25 (1.18–1.33)	<0.001	1.18 (1.10–1.27)	<0.001
SAT volume (per SD)	1.09 (1.03–1.16)	0.002	0.77 (0.71–0.83)	<0.001
VAT/SAT volume	1.09 (1.04–1.14)	<0.001	1.06 (1.00–1.13)	0.064

Table 2. Sex-specific associations of abdominal fat compartments with CKD risk. *Adjusted for age, hypertension and diabetes. *OR* odds ratio, *CI* confidence interval, *SD* standard deviation, *VAT* visceral adipose tissue, *SAT* subcutaneous adipose tissue.

ratio and CKD risk in both sexes, supporting the validity of the linear modeling approach used in the main analyses (Supplementary Figure S1).

Sensitivity analysis

To evaluate the robustness of the findings, two sensitivity analyses were performed.

First, after excluding participants taking anti-hypertensive, anti-diabetic, or lipid-lowering medications at baseline, the associations between VAT, SAT, and CKD risk remained significant and directionally consistent with the main analyses (Supplementary Table 1).

Variable	Male			Female		
	range, (cm ³)	OR (95% CI)	P value	range, (cm ³)	OR (95% CI)	P value
VAT volume (categorical)						
Quartile 1	[191–1,075]	(reference)		[89–379]	(reference)	
Quartile 2	[1,075–1,492]	1.27 (1.12–1.43)	<0.001	[379–628]	0.61 (0.52–0.72)	<0.001
Quartile 3	[1,492–1,925]	1.66 (1.47–1.88)	<0.001	[628–920]	1.09 (0.94–1.27)	0.271
Quartile 4	[1,925–2,895]	1.77 (1.57–2.00)	<0.001	[920–1,659]	1.29 (1.10–1.52)	0.002
SAT volume (categorical)						
Quartile 1	[374–1,010]	(reference)		[570–1,262]	(reference)	
Quartile 2	[1,010–1,254]	1.11 (0.99–1.24)	0.080	[1,262–1,565]	0.67 (0.57–0.78)	<0.001
Quartile 3	[1,254–1,540]	1.00 (0.89–1.13)	0.938	[1,565–1,896]	0.97 (0.84–1.12)	0.671
Quartile 4	[1,540–2,320]	1.29 (1.15–1.45)	<0.001	[1,896–2,758]	0.39 (0.33–0.45)	<0.001
VAT/SAT volume (categorical)						
Quartile 1	[0.15–0.88]	(reference)		[0.08–0.26]	(reference)	
Quartile 2	[0.88–1.14]	1.57 (1.38–1.79)	<0.001	[0.26–0.38]	0.39 (0.32–0.47)	<0.001
Quartile 3	[1.14–1.44]	1.44 (1.26–1.64)	<0.001	[0.38–0.53]	0.24 (0.20–0.29)	<0.001
Quartile 4	[1.44–4.33]	1.88 (1.66–2.14)	<0.001	[0.53–2.29]	0.65 (0.55–0.77)	<0.001

Table 3. Multivariable logistic regression analyses of CKD risk across quantiles of VAT and SAT. CKD chronic kidney disease, OR odds ratio, CI confidence interval, VAT visceral adipose tissue, SAT subcutaneous adipose tissue. Adjusted for age, hypertension and diabetes.

Second, eGFR was recalculated using the CKD-EPI equation instead of the MDRD formula. The associations between abdominal fat compartments and CKD risk were largely unchanged, confirming the robustness of the results (Supplementary Table 2).

Discussion

This study demonstrated that VAT volume, assessed using a deep neural network architecture applied to abdominal CT scans, is significantly associated with a higher risk of CKD in both males and females. The volume of SAT was significantly associated with a decreased risk of CKD in females but an increased risk in males. Moreover, the VAT/SAT ratio was significantly associated with an increased risk of CKD in males. These findings highlight the relationship between body composition, particularly visceral fat and CKD risk, while also emphasizing sex-specific differences in the impact of subcutaneous fat.

Our finding that volumetrically quantified VAT volume was strongly associated with CKD risk extends prior research that relied on limited two-dimensional CT slices. A systematic review has established obesity as an important risk factor for CKD¹⁹. Although BMI and WC are commonly used to assess obesity, these metrics offer only a crude approximation of body composition and fail to accurately reflect the distribution of fat and muscle, particularly in the abdominal region. Our previous research demonstrated that VAT as an independent risk factor for CKD incidence²⁰. In that study, abdominal fat areas were manually traced on two-dimensional CT images at the L3 lumbar level, following established methodologies^{21,22}. However, this manual approach is inherently limited by inter- and intra-observer variability, which can introduce measurement bias and reduce reproducibility.

In the present study, we addressed these limitations by applying an automated deep-learning-based segmentation algorithm, which eliminates observer-dependent variability and ensures consistent quantification across large datasets. A major strength of this approach is the volumetric assessment of abdominal fat, providing a three-dimensional representation that more accurately reflects an individual's visceral and subcutaneous adiposity compared with traditional 2D planimetric methods. This high-resolution volumetric information enhances our ability to elucidate the distinct contributions of VAT and SAT to metabolic health and CKD risk, recognizing their fundamentally different pathophysiologic roles.

VAT plays a central role in obesity-related metabolic diseases, unlike SAT, which is often considered benign or even protective^{23,24}. This study observed a positive association between VAT and CKD prevalence, while SAT demonstrated a negative association in females. The increased CKD risk associated with VAT volume might be driven by several mechanisms. Initially, VAT contributes to heightened renal hemodynamic stress, including sympathetic nervous system overactivation^{25,26}, elevated renin-angiotensin-aldosterone system activity^{27,28}, and sodium retention¹⁸. Furthermore, increased visceral fat frequently coincides with perirenal fat accumulation, resulting in elevated intrarenal pressure. This, in turn, leads to decreased renal blood flow, enhanced sodium reabsorption, and renin-angiotensin-aldosterone system activation, ultimately culminating in glomerular hyperfiltration^{29,30}. This process, a hallmark of early renal impairment, places a significant burden on the kidneys, thereby hastening CKD progression. Moreover, the metabolically active nature of visceral fat contributes to adipokine dysregulation, oxidative stress, systemic inflammation, and insulin resistance, all of which negatively impact renal function and accelerate CKD progression^{31,32}.

This study identified sex-specific differences in the relationship between SAT and CKD. While VAT was consistently associated with an increased CKD risk in both sexes, SAT exhibited a protective effect in females

but was positively linked to CKD risk in males. This finding aligns with previous research emphasizing the significance of sex differences in the complex interaction between body fat and CKD³³, likely influenced by the well-documented role of sex hormones in fat distribution³⁴. These observed differences may be attributed to the generally higher proportion of visceral fat in males and greater subcutaneous fat accumulation in females³⁵, which could impact the association between SAT and CKD risk.

Supporting this, recent studies on Korean populations have demonstrated sex-specific variations in glomerular hyperfiltration—an early indicator of renal dysfunction—in the context of obesity³⁶. These studies found that glomerular hyperfiltration significantly increased mortality risk in males but not in females. Researchers suggested that this discrepancy may be linked to the well-established observation that females with CKD tend to have lower cardiovascular and overall mortality rates compared to their male counterparts³⁷. However, the precise mechanisms underlying these sex-specific differences in the relationship between body fat distribution and CKD remain unclear, warranting further investigation.

This study has several limitations that should be considered. First, as our study population consisted exclusively of Asians, the generalizability of our findings to other ethnicities or racial groups is limited. Since Asian populations tend to have a lower prevalence of obesity and CKD risk factors compared to other groups, our results may reflect a relatively healthier cohort than those examined in previous studies. Second, we were unable to establish a causal relationship between body composition and CKD, as our analysis was limited to assessing associations. Third, the absence of repeated serum creatinine measurements prevented the definitive confirmation of chronic kidney dysfunction. Consequently, our CKD definition may have inadvertently included transient reductions in the eGFR, potentially introducing misclassification bias. Furthermore, because our study population consisted of individuals undergoing health screening CT scans, patients with impaired renal function may have been underrepresented. This selection bias could lead to a lower CKD prevalence compared with the general population and may further limit the generalizability of our findings. Moreover, this study was conducted at a single center, which may limit external validity. Institutional characteristics and patient demographics might not fully represent broader populations. Future multicenter studies with more diverse cohorts are warranted to confirm and extend our findings. Lastly, while CT-based body composition assessment presents challenges due to radiation exposure and the potential use of contrast agents, this study mitigated such concerns by utilizing non-contrast CT images.

This study highlights a significant association between VAT volume, assessed through deep learning-based analysis of abdominal CT scans, and an increased risk of CKD in both males and females. Meanwhile, SAT volume showed a sex-specific relationship with CKD risk, demonstrating a protective effect in females but a positive association in males. AI-driven analysis of abdominal CT images presents a promising tool for CKD risk assessment; however, further studies are required to clarify the underlying mechanisms behind the observed sex-specific differences in the link between body composition and CKD.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 1 May 2025; Accepted: 24 November 2025

Published online: 04 March 2026

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Author contributions

N.J.H. was responsible for conceptualization and methodology. G.E.C. and J.W.Y. curated the data. G.E.C., and N.J.H. visualized and wrote the original draft. H.K. performed the formal analysis. Y.M.H. and S.C. were responsible for reviewing and editing the manuscript. N.J.H. is the guarantor of this work and, as such, had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. All authors reviewed and edited the manuscript.

Funding

This research was supported by a grant from the Medical Data-driven Hospital Support Project through the Korea Health Information Service, funded by the Ministry of Health & Welfare, Republic of Korea.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-30244-6>.

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